ST MARY'S DSG, KLOOF

GRADE: 10 **17 October 2025**

PHYSICAL SCIENCE CYCLE TEST
KINEMATICS AND ELECTRICITY

TIME: 55 MINS (incl. reading time) **TOTAL: 63**
ASSESSOR: T. Kirsten **MODERATORS: K. Bolton and G. Main**

NAME:_____

SCIENCE TEACHER:_____

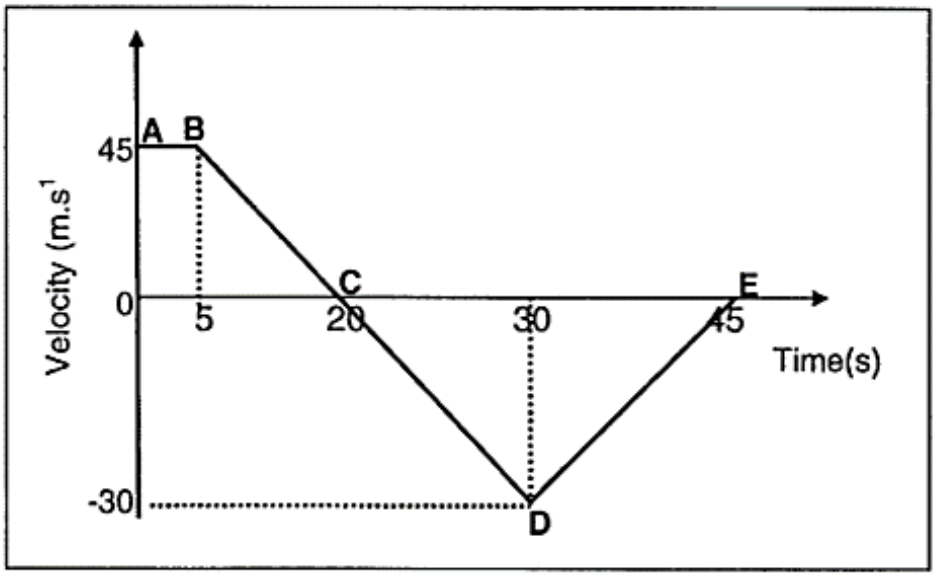
PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 8 pages. A data sheet is provided.
2. Answer on the question paper in the spaces.
3. Write your name and your science teacher's name on your question paper.

SECTION A: KINEMATICS

QUESTION 1

The velocity-time graph of a train initially moving **in a SOUTHERLY direction** is shown below. The graph is NOT drawn to scale.



1.1 Describe the motion of the train for the following time intervals, in terms of velocity and acceleration. (8)

1.1.1 B to C

1.1.2 C to D

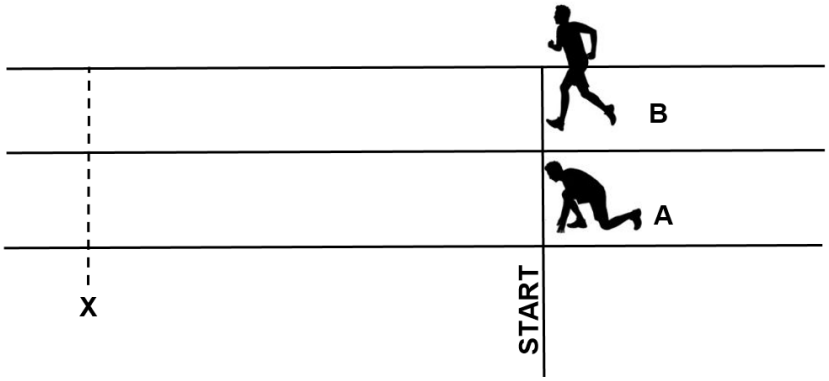
1.2 Without the use of equations of motion, calculate the total displacement for the time interval A-E. (6)

1.3 Sketch an acceleration vs time graph for the train over the entire journey. Indicate all relevant time values. Acceleration values are not necessary. (4)

[18]

QUESTION 2

Athletes **A** and **B** are training on a straight, level track.



- Athlete **A** starts from rest from the position shown.
- Athlete **A** accelerates to the left at 0,4 $m \cdot s^{-2}$.
- Athlete **B** is running to the left at a constant velocity of 2 $m \cdot s^{-1}$.
- At the instant that athlete **A** starts, athlete **B** passes athlete **A.**

2.1 Determine the time at which **A** and **B** are in the same position **X**. (4)

2.2 Determine the magnitude of the displacement of the athlete **A** at point **X**, taking the start as the reference position. (3)

- Athlete **A** accelerates at 0,4 $m \cdot s^{-2}$ for 12 s.
- She then continues to run to the left at constant velocity for a further 10 s.
- She then comes to rest over another 10 s.

2.3 In the space below, draw a position-time sketch graph for athlete **A** for the full 32 s of her run. Take left as positive and use the start as the reference point.
No values need to be shown on the y-axis **but you should label times on the x- axis.** (5)

Athlete **B** decides to slow down when passing point **X** and comes to rest during a time of 7 s.

2.4 Define *acceleration*. (2)

2.5 Calculate the acceleration of athlete **B** during this time period. (4)

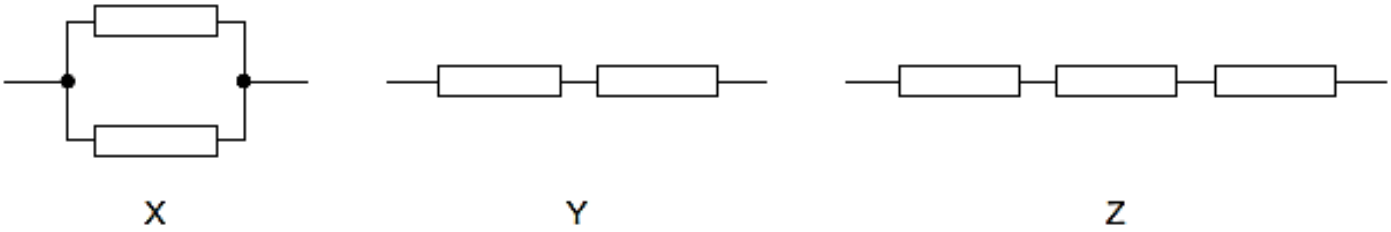
[18]

SECTION B: ELECTRICITY

QUESTION 3 MULTIPLE CHOICE Circle the most correct letter

3.1 Identical resistors are connected together to form arrangements X, Y and Z.

What is the correct order of the total resistances of the arrangements from the **largest to the smallest?**



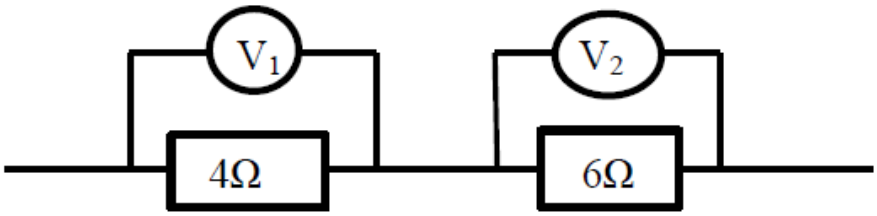
A X → Y → Z

B Y → X → Z

C Z → X → Y

D Z → Y → X

3.2 Consider the diagram below:



If a current is flowing through both resistors;

A V_1 will have a higher reading than V_2

B V_2 will have a higher reading than V_1

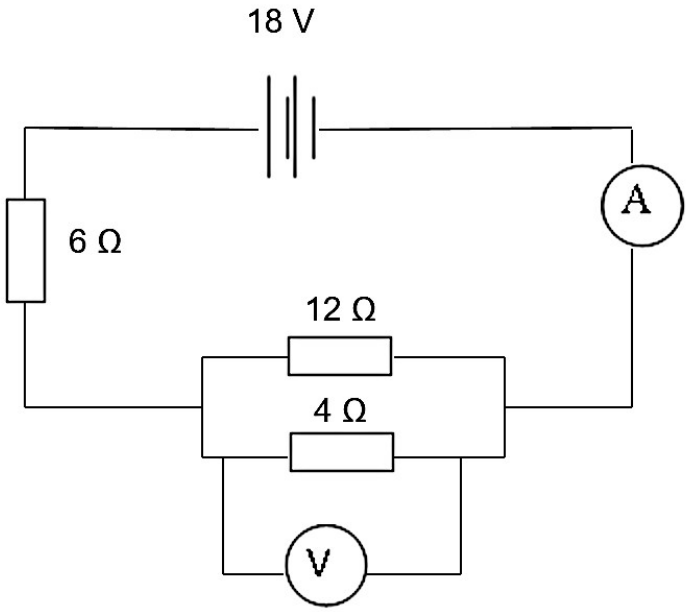
C the reading on V_1 and V_2 will be the same

D potential difference is inversely proportional to resistance.

2 x 2 = [4]

QUESTION 4

A 18 V battery is connected in a circuit as shown in the diagram below.



4.1. Define *resistance*. (2)

4.2 Calculate:

4.2.1 the resistance of the parallel combination (3)

4.2.2 the total resistance of the given circuit (2)

4.3 Define *current*. (2)

4.4 Calculate:

4.4.1 the reading on the ammeter (2)

4.4.2 the time it takes for 17 C of charge to pass a point in the above circuit. (3)

4.5 Define *potential difference*. (2)

4.6 Calculate:

4.6.1 the reading on the voltmeter (3)

4.6.2 the work done in moving 5 Coulombs of charge through the 6 Ω resistor. (4)

[23]